Minseo Kim

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Research Interests

AI Systems, Efficient Inference across Modern Architectures and Workloads

Education

- **University of California, Berkeley** — *Aug 2026 – Present*
  Ph.D. in Computer Science — Berkeley, CA, USA
  ◦ Advisor: Prof. Kurt Keutzer
- **Seoul National University** — *Mar 2020 – Aug 2026*
  *B.S. in Computer Science and Engineering, **Summa cum laude** (GPA: 4.03/4.30)* — Seoul, South Korea
  ◦ Military service in the Republic of Korea Air Force (Oct 2021 – Jul 2023)
- **University of California, Berkeley** — *Jun 2025 – Aug 2025*
  Summer Session Visiting Student (GPA: 4.0/4.0) — Berkeley, CA, USA
- **Korea Minjok Leadership Academy** — *Mar 2017 – Feb 2020*
  *Secondary Education, **First-class honors** in STEM (1st / 73)* — Gangwon-do, South Korea

Research Experience

- **Berkeley Artificial Intelligence Research (BAIR) Lab [🌐]** — *Aug 2026 – Present; Jun 2025 – Jan 2026*
  Ph.D. Student (Advisor: Prof. Kurt Keutzer) — Berkeley, CA, USA
  Undergraduate Researcher
- **SNU Architecture and Code Optimization Lab [🌐]** — *Jan 2025 – Feb 2025*
  Undergraduate Researcher (Advisor: Prof. Jae W. Lee) — Seoul, South Korea

Work Experience

- **FuriosaAI [🌐]** — *Jan 2026 – Jul 2026*
  AI Algorithm Research Intern — Seoul, South Korea
- **Deeping Source Inc. [🌐]** — *Jul 2024 – Aug 2024*
  ML Engineering Intern — Seoul, South Korea

Publications

* Equal contribution † Co-corresponding authors

[6] EfficientRollout: System-Aware Self-Speculative Decoding for RL Rollouts. [Paper] [Code]
Minseo Kim*, Minjae Lee*, Seunghyuk Oh, Kevin Galim, Donghoon Kim, Coleman Hooper, Harman Singh, Amir Gholami, Hyung Il Koo, Wonjun Kang
NeurIPS 2026; ICML 2026 AdaptFM Workshop

[5] AsyncOPD: How Stale Can On-Policy Distillation Be? [Paper] [Code]
Wonjun Kang*, Kevin Galim*, Seunghyuk Oh, Minjun Kang, Sanghyun Park, Donghoon Kim, Minjae Lee, **Minseo Kim**, Rishabh Tiwari, Yuchen Zeng, Hyung Il Koo, Kangwook Lee
NeurIPS 2026; ICML 2026 DEMO Workshop

[4] MAGE: All-[MASK] Block Already Knows Where to Look in Block Diffusion LLM. [Paper]
Omin Kwon, Yeonjae Kim, Doyeon Kim, **Minseo Kim**, Yeonhong Park, Jae W. Lee
NeurIPS 2026; ICML 2026 AdaptFM Workshop **Best Paper Award**

[3] CDLM: Consistency Diffusion Language Models for Faster Sampling. [Paper] [Code]
Minseo Kim, Chenfeng Xu, Coleman Hooper, Harman Singh, Ben Athiwaratkun, Ce Zhang, Kurt Keutzer, Amir Gholami
MLSys 2026

[2] KRETA: A Benchmark for Korean Reading and Reasoning in Text-Rich VQA Attuned to Diverse Visual Contexts. [Paper] [Code] [Dataset]
Taebaek Hwang*, **Minseo Kim***, Gisang Lee, Seonuk Kim, Hyunjun Eun
EMNLP 2025

[1] InstANNS: Scalable Approximate Nearest Neighbor Search via Cost-Efficient In-Storage Processing. [Paper]
Bonggeun Sim, Yushin Kim, **Minseo Kim**, Yeonhong Park, Jae W. Lee
CIKM 2025

Preprints & Workshop Papers

[6] MetaCurate: Evolving Curation Filters for High-Quality Agentic Coding Data. [Paper] [Code]
Donghoon Kim*, Seunghyuk Oh*, Kevin Galim, **Minseo Kim**, Minjae Lee, Younghyun Kim, Wooseok Seo, Hyung Il Koo, Wonjun Kang
Preprint, 2026

[5] Elicitation or Expansion? Demystifying Gains from Agent Harness Optimization. [Paper] [Code]
Younghyun Kim*, Wooseok Seo*, **Minseo Kim**, Seunghyuk Oh, Minjae Lee, Donghoon Kim, Hyung Il Koo, Jaehyung Kim†, Wonjun Kang†
Preprint, 2026; Conference submission under review

[4] SketchSSM: Write to the Full State, Read from a Compact Sketch. [Paper]
Omin Kwon, JoongWon Shin, **Minseo Kim**, Kurt Keutzer, Sehoon Kim†, Jae W. Lee†
NeurIPS 2026 LCFM Workshop; Conference submission under review

[3] LLM Inference in a Flash! [Paper]
Sebastian Zhao*, **Minseo Kim***, Coleman Hooper*, Luca Manolache, Michael W. Mahoney, Yakun Sophia Shao, Kurt Keutzer, Amir Gholami
ISCA 2026 MLArchSys Workshop **Oral**

[2] CATS: Inference-aligned SFT for Diffusion LLMs via Context-sensitivity Aware Trajectory Sampling. [Paper]
Seunghyuk Oh, Minjae Lee, Kevin Galim, **Minseo Kim**, Hyung Il Koo, Wonjun Kang, Hanbaek Lyu, Kangwook Lee
ICLR 2026 DeLTa Workshop

[1] Beyond Next-Token Prediction: A Performance Characterization of Diffusion versus Autoregressive Language Models. [Paper]
Minseo Kim, Coleman Hooper, Aditya Tomar, Chenfeng Xu, Mehrdad Farajtabar, Michael W. Mahoney, Kurt Keutzer, Amir Gholami
Preprint, 2025

Awards & Achievements

- **2025 AI Chip Contest – Grand Prize (Top Award)** *Dec 2025*
- **Ministry of Science and ICT Award - Whitehat School Outstanding Graduate** *Mar 2024*
- **Vulnerability Disclosures** *2023*
  - CVE-2024-11510, CVE-2024-11511, CVE-2024-11512.

Scholarships

- **Gaheonsindo Foundation Scholarship (Full Undergraduate Tuition)** *2024 – 2026*
- **SNU Specialized Semiconductor College Scholarship** *2024 – 2026*

Talks & Leadership

- **Diffusion Language Model Seminar @ Cerebras** (Virtual) *Jan 2026*
- **President @ Guardian** (SNU CSE Security Research Club) *Aug 2024 – Aug 2025*
- **Windows SW Vulnerability Analysis Seminar @ 28th Winter Hacking Camp** *Feb 2024*

Language Proficiency

Korean (Native), English (Fluent)

References

University of California, Berkeley Prof. Kurt Keutzer (Ph.D. Advisor)
FuriosaAI Dr. Wonjun Kang (Internship Mentor)
Seoul National University Prof. Jae W. Lee (Undergraduate Advisor)